NPC – Unit 3 Assignment

Topic: Configure a DNS Server and Web Server Using Cisco Packet Tracer

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Submitted By

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1. AIM

To design and configure a network in Cisco Packet Tracer consisting of PCs, a switch, a DNS server, and a web server. The aim is to assign IP addresses to all devices, enable HTTP service on the web server, configure DNS to map a domain name to the web server's IP address, and verify connectivity and website access.

2. OBJECTIVES

1. To create a network topology using Cisco Packet Tracer.
2. To assign static IPv4 addresses to all devices.
3. To configure a web server and enable HTTP service.
4. To configure a DNS server with an A record mapping a domain name to the web server.
5. To configure client PCs to use the DNS server.
6. To test network connectivity using the ping command.
7. To access the website using its domain name in a web browser.

3. SOFTWARE AND HARDWARE REQUIREMENTS

Software: Cisco Packet Tracer

Devices used:

- 2 Server-PT devices
- 2 PC-PT devices
- 1 Cisco 2960-24TT switch
- Copper Straight-Through Ethernet cables

4. NETWORK TOPOLOGY

A star topology was created in Cisco Packet Tracer. Both servers and both client PCs are connected to a central switch using Copper Straight-Through cables.

Server0 is configured as the DNS server, Server1 is configured as the web server, and PC0 and PC1 act as client devices.

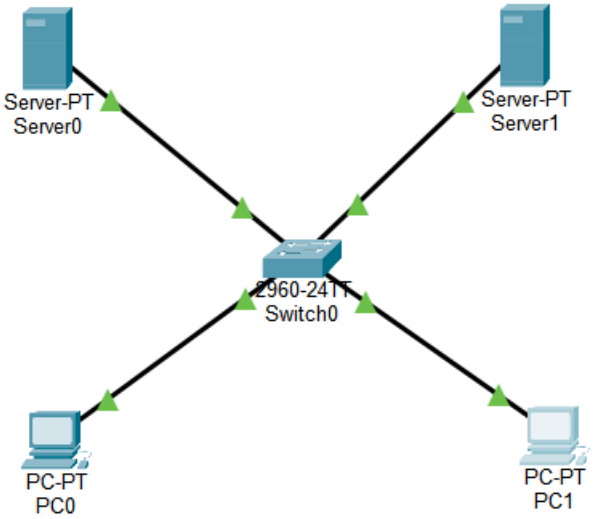


Figure 1: Network topology for DNS and web server configuration.

5. IP ADDRESSING CONFIGURATION

The following static IPv4 addresses were assigned to the network devices.

Device	Role	IPv4 Address	Subnet Mask	DNS Server
Server0	DNS Server	192.168.1.2	255.255.255.0	192.168.1.2
Server1	Web Server	192.168.1.3	255.255.255.0	192.168.1.2

PC0	Client	192.168.1.10	255.255.255.0	192.168.1.2
PC1	Client	192.168.1.11	255.255.255.0	192.168.1.2

The default gateway was left blank because all devices belong to the same local subnet and no router is required for this configuration.

5.1 Configuring Server0

The following steps were performed to configure Server0:

1. Opened Server0 in Cisco Packet Tracer.
2. Selected the Desktop tab.
3. Opened IP Configuration.
4. Entered the IPv4 address as 192.168.1.2.
5. Set the subnet mask to 255.255.255.0.
6. Set the DNS server address to 192.168.1.2.

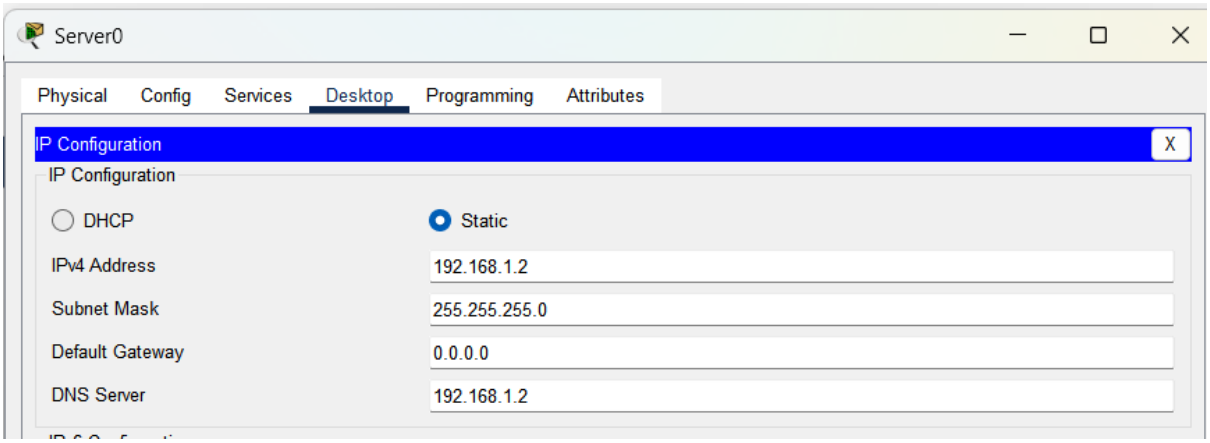


Figure 2: IP configuration of the DNS server.

5.2 Configuring Server1

The following steps were performed to configure Server1:

1. Opened Server1.

2. Selected Desktop → IP Configuration.
3. Entered the IPv4 address as 192.168.1.3.
4. Set the subnet mask to 255.255.255.0.
5. Set the DNS server address to 192.168.1.2.

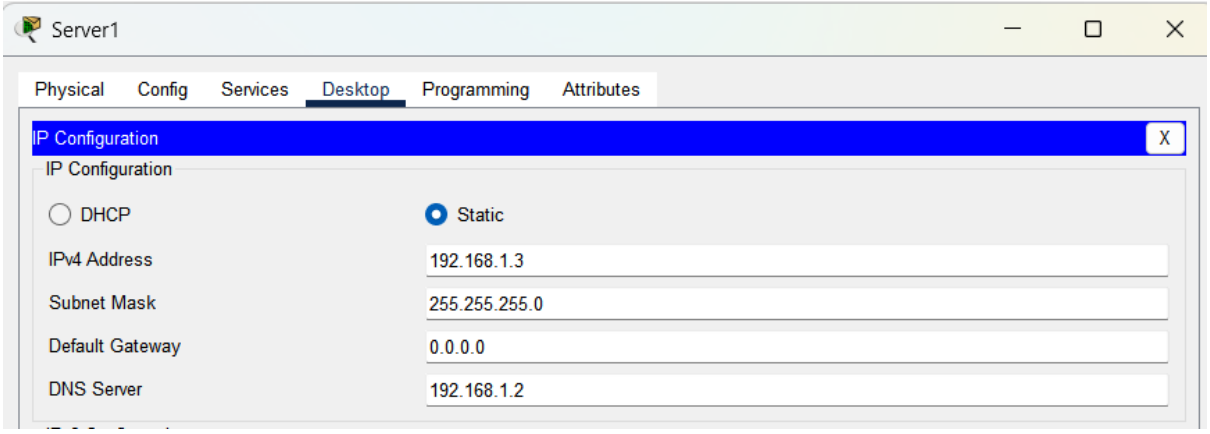


Figure 3: IP configuration of the web server.

5.3 Configuring PC0 and PC1

PC0 and PC1 were configured with static IPv4 addresses.

For PC0:

- IPv4 Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- DNS Server: 192.168.1.2

For PC1:

- IPv4 Address: 192.168.1.11
- Subnet Mask: 255.255.255.0
- DNS Server: 192.168.1.2

The configuration was performed through Desktop → IP Configuration on each PC.

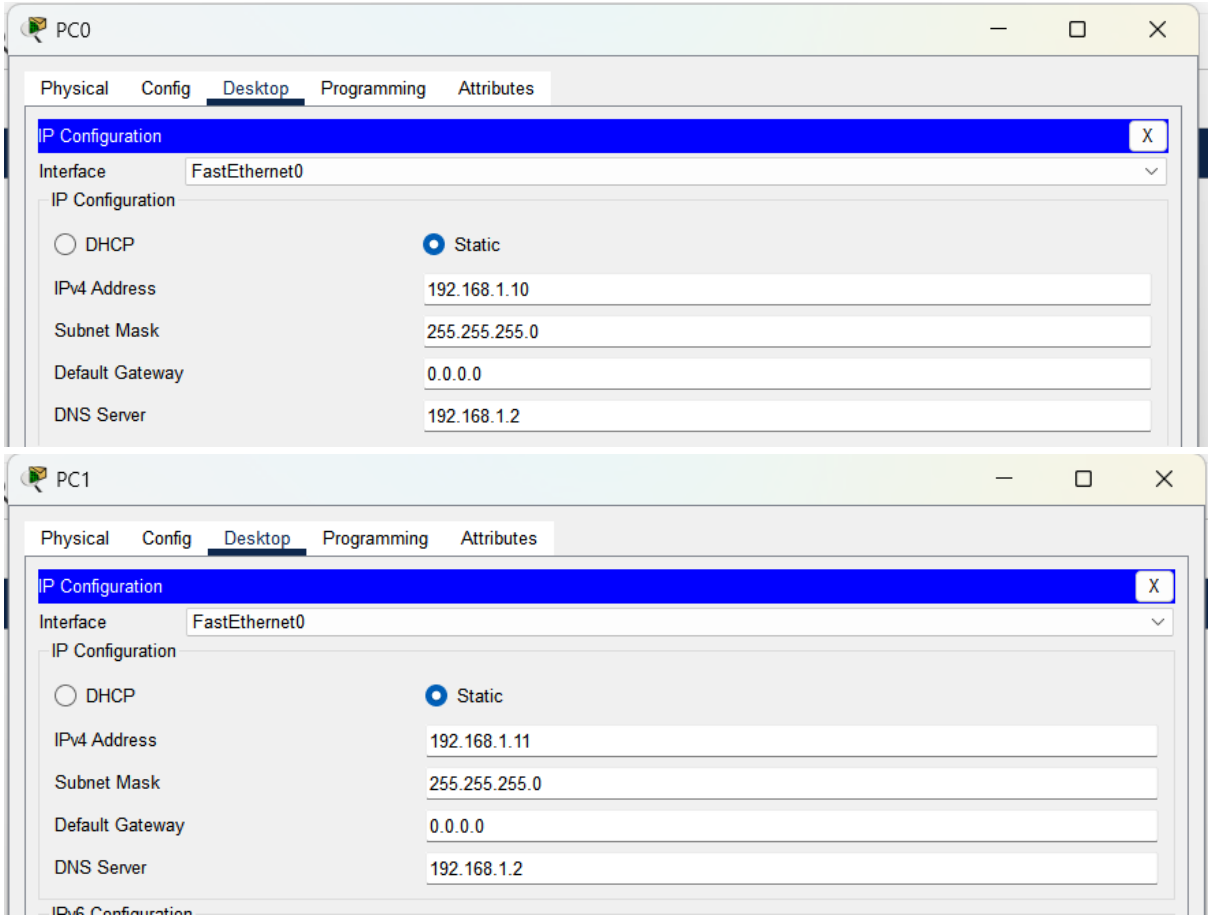


Figure 4: IP configuration of the client PC.

6. CONFIGURATION OF THE WEB SERVER

Server1 was configured to host a website using HTTP.

The following procedure was followed:

1. Opened Server1.
2. Selected the Services tab.
3. Selected HTTP from the services menu.
4. Enabled HTTP by setting the service to On.
5. Retained the default index.html webpage or edited it to display a custom welcome message.

The web server was assigned the IP address 192.168.1.3. Therefore, clients can access the website directly using its IP address or through a domain name after DNS configuration.

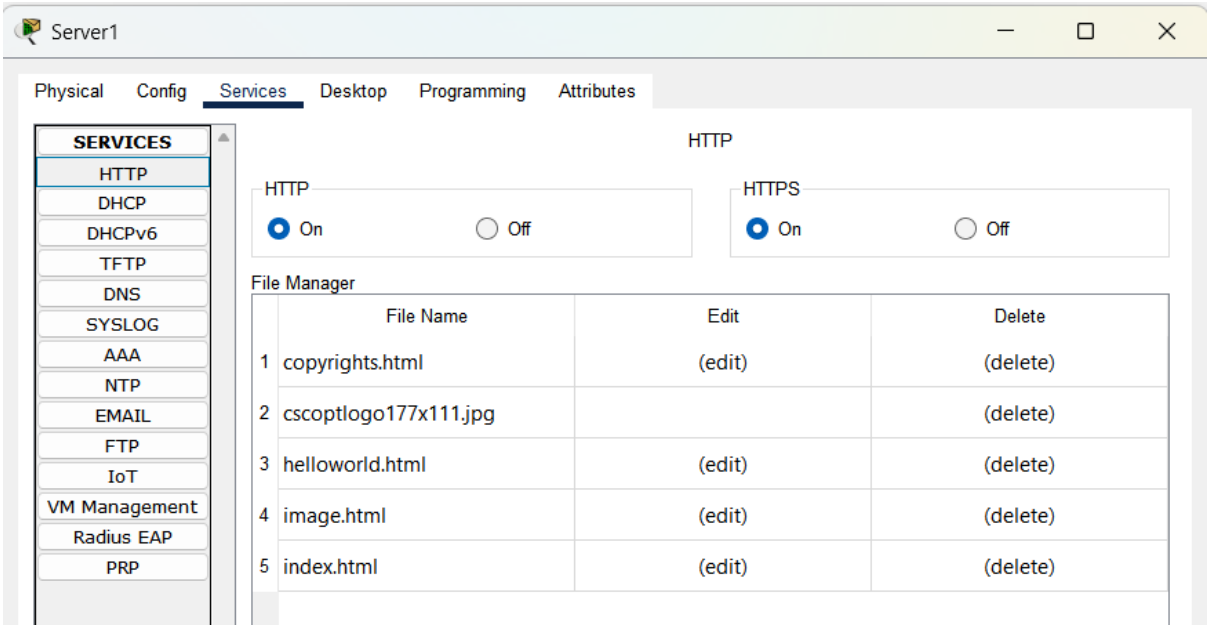


Figure 5: HTTP service enabled on the web server.

7. CONFIGURATION OF THE DNS SERVER

The DNS server translates a domain name into the corresponding IP address of the web server.

Server0 was configured to provide DNS services using the following steps:

1. Opened Server0.
2. Selected the Services tab.
3. Selected DNS from the services menu.
4. Turned the DNS service On.
5. Added an A record mapping the domain name to the web server's IP address.

The DNS record was configured as follows:

Parameter	Value
DNS Service	On
Record Name	www.mimi.com
Record Type	A Record
IP Address	192.168.1.3

The A record maps the domain name www.mimi.com to the IPv4 address 192.168.1.3, which belongs to Server1.

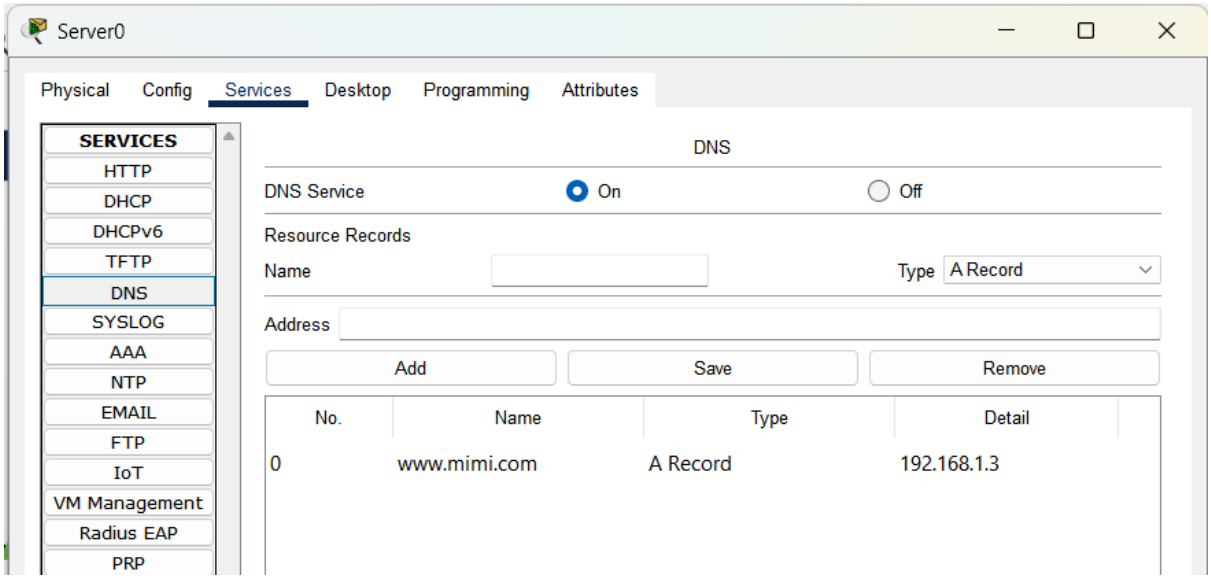


Figure 6: DNS configuration and domain name mapping.

8. TESTING AND VERIFICATION

After configuring the devices and services, connectivity and DNS resolution were tested from PC0.

8.1 Testing Connectivity to the DNS Server

The following command was executed in PC0's Command Prompt:

ping 192.168.1.2

The command was used to verify connectivity between PC0 and the DNS server.

Result: Four packets were sent and four packets were received, with 0% packet loss.

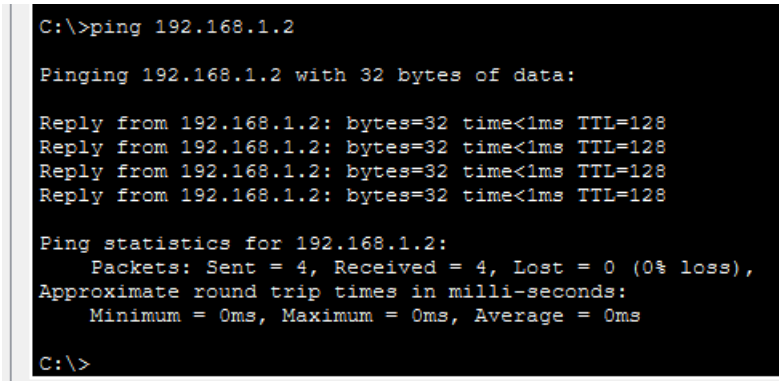


Figure 7: Successful connectivity test to the DNS server.

8.2 Testing Connectivity to the Web Server

The following command was executed:

ping 192.168.1.3

This test verified connectivity between PC0 and the web server.

Result: Four packets were sent and four packets were received, with 0% packet loss.

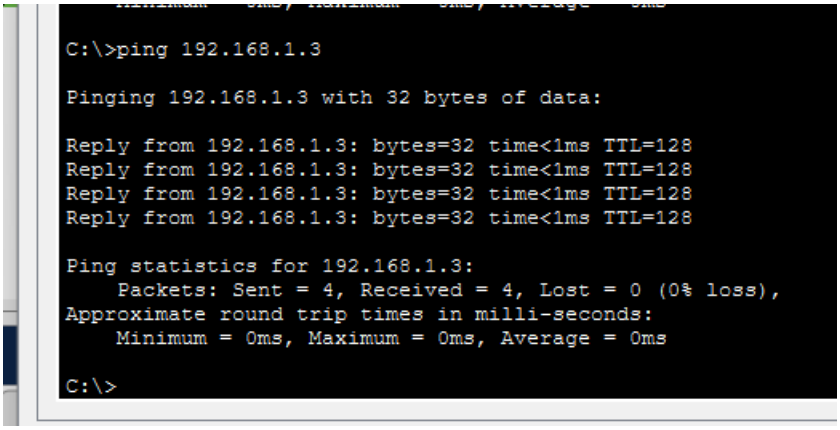


Figure 8: Successful connectivity test to the web server.

8.3 Testing DNS Name Resolution

After configuring the DNS service and adding the A record, the following command was executed:

ping www.mimi.com

The DNS server resolved the domain name www.mimi.com to the IP address 192.168.1.3. The ping command then sent packets to the resolved address.

Result: The hostname resolved successfully to 192.168.1.3. Four packets were received, and the test showed 0% packet loss.

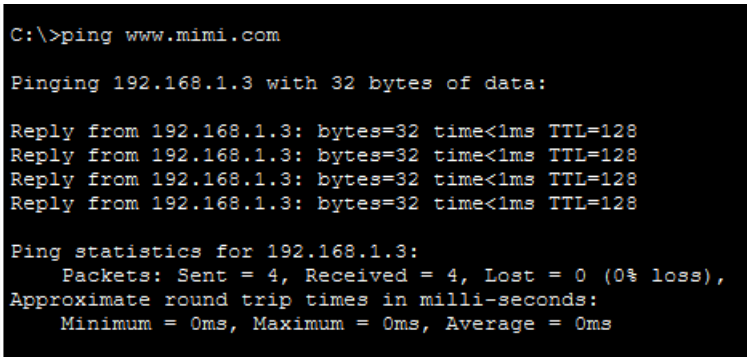


Figure 9: Successful DNS name resolution and connectivity test.

8.4 Accessing the Website Through the Domain Name

The final verification was performed through the web browser on PC0.

Procedure:

1. Opened PC0.
2. Selected Desktop → Web Browser.
3. Entered the URL http://www.mimi.com.
4. Clicked Go to access the website hosted on Server1.

Expected result: The browser displays the webpage hosted on Server1, confirming that the domain name can be used to access the website.

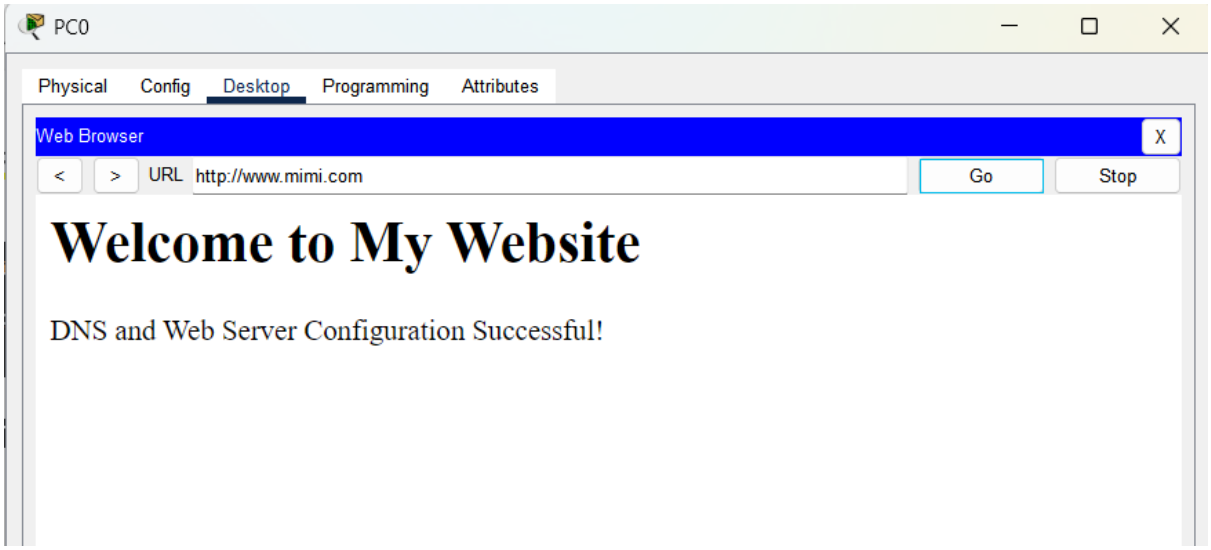


Figure 10: Website accessed using its configured domain name.

9. TEST RESULTS

The following table summarizes the tests performed during the practical.

Test	Target	Result
IP connectivity	192.168.1.2	Successful; 0% packet loss
IP connectivity	192.168.1.3	Successful; 0% packet loss
DNS resolution	www.mimi.com	Resolved to 192.168.1.3
Website access	http://www.mimi.com	To be verified and documented with the browser screenshot

The connectivity tests confirmed that PC0 can communicate with both servers. The DNS test confirmed that the configured domain name resolves to the web server's IP address. The final browser test documents successful HTTP access through the domain name.

10. CONCLUSION

The DNS and web server network was configured successfully in Cisco Packet Tracer. Static IPv4 addresses were assigned to the servers and client PCs, and connectivity between PC0 and both servers was verified using the ping command.

The DNS service was configured with an A record mapping www.mimi.com to the web server's IP address, 192.168.1.3. The successful DNS resolution test confirmed that the domain name was translated into the correct IP address.

The practical demonstrates the role of DNS in resolving domain names and HTTP in delivering webpages to client devices. The final browser screenshot should be included to document successful access to the hosted website through its domain name.

11. LEARNING OUTCOMES

Through this practical, the following concepts were understood:

1. Creating a LAN topology in Cisco Packet Tracer.
2. Assigning static IPv4 addresses and subnet masks.
3. Configuring DNS and HTTP services on server devices.
4. Creating an A record to map a domain name to an IPv4 address.
5. Testing connectivity using the ping command.

6. Understanding how DNS name resolution enables users to access websites using domain names rather than IP addresses.

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